

Data Preprocessing

To solve a business problem using analytics, you need to have historical data to come up with actionable insights. Data is the key — the better the data, the more insights you can get out of it.

Typically, data comes from various sources and your first job as a data analyst is to procure the data from them. In this module, you have learnt about various sources of data and how to source data from public and private sources.

Public Data

A large amount of data collected by the government or other public agencies is made public for the purposes of research. Such data sets do not require special permission for access and are therefore called public data. Public data is available on the internet on various platforms. A lot of data sets are available for direct analysis, whereas some of the data have to be manually extracted and converted into a format that is fit for analysis.

Private Data

Private data is that which is sensitive to organisations and is thus not available in the public domain. Banking, telecom, retail, and media are some of the key private sectors that rely heavily on data to make decisions. A large number of organisations seek to leverage data analytics to make crucial decisions. As organisations become customer-centric, they utilise insights from data to enhance customer experience, while also optimising their daily processes.

There are various types of quality issues when it comes to data, and that's why data cleaning is one of the most time-consuming steps of data analysis. For example, there could be formatting errors (e.g. rows and columns are merged), missing values, repeated rows, spelling inconsistencies etc. These issues could make it difficult to analyse data and could lead to errors or irrelevant results. Thus, these issues need to be corrected before data is analysed.

Though data cleaning is often done in a somewhat haphazard way and it is too difficult to define a 'single structured process', we will study data cleaning in the following steps:

1. Fixing rows and columns
2. Handling missing values
3. Handling outliers
4. Standardising values
5. Fixing invalid values
6. Filtering data

Checklist for fixing rows:

- **Delete summary rows:** Total, Subtotal rows
- **Delete incorrect rows:** Header rows, Footer rows

- **Delete extra rows:** Column number, indicators, Blank rows, Page No.

Checklist for fixing columns

- **Merge columns** for creating unique identifiers if needed: e.g. Merge State, City into Full address
- **Split columns for more data:** Split address to get State and City to analyse each separately
- **Add column names:** Add column names if missing
- **Rename columns consistently:** Abbreviations, encoded columns
- **Delete columns:** Delete unnecessary columns
- **Align misaligned columns:** Data set may have shifted columns

Checklist for handling missing values

- **Set values as missing values:** Identify values that indicate missing data and yet are not recognised by the software as such, e.g treat blank strings, "NA", "XX", "999", etc. as missing.
- **Adding is good, exaggerating is bad:** You should try to get information from reliable external sources as much as possible, but if you can't, then it is better to keep missing values as such rather than exaggerating the existing rows/columns.
- **Delete rows, columns:** Rows could be deleted if the missing values are insignificant in number, as this would not impact the analysis. Columns could be removed if the missing values are quite significant in number.
- **Fill partial missing values using business judgement:** Missing time zone, century, etc. These values are easily identifiable.

Checklist for standardising values:

- **Standardise units:** Ensure all observations under a variable have a common and consistent unit, e.g. convert lbs to kgs, miles/hr to km/hr, etc.
- **Scale values if required:** Make sure the observations under a variable have a common scale
- **Standardise precision** for better presentation of data, e.g. 4.5312341 kgs to 4.53 kgs.
- **Remove outliers:** Remove high and low values that would disproportionately affect the results of your analysis.

Checklist for handling invalid values

A data set can contain invalid values in various forms. Some of the values could be truly invalid, e.g. a string "tr8ml" in a variable containing mobile numbers would make no sense and hence would be better removed. Similarly, a height of 11 ft would be an invalid value in a set containing heights of children.

On the other hand, some invalid values can be corrected. e.g. a numeric value with a data type of string could be converted to its original numeric type. Issues might arise due to R misinterpreting the encoding of a file, thus showing junk characters where there were valid characters. This could be corrected by correctly specifying the encoding or converting the data set to the accurate format before importing.

- **Encode unicode properly:** In case the data is being read as junk characters, try to change encoding, e.g. CP1252 instead of UTF-8.
- **Convert incorrect data types:** Correct the incorrect data types to the correct data types for ease of analysis.

E.g., if numeric values are stored as strings, it would not be possible to calculate metrics such as mean, median, etc. Some of the common data type corrections are — string to number: "12,300" to "12300"; string to date: "2013-Aug" to "2013/08"; number to string: "PIN Code 110001" to "110001"; etc.

- **Correct values that go beyond range:** If some of the values are beyond logical range, e.g. temperature less than -273° C (0° K), you would need to correct them as required. A close look would help you check if there is scope for correction, or if the value needs to be removed.
- **Correct values not in the list:** Remove values that don't belong to a list. E.g. In a data set containing blood groups of individuals, strings "E" or "F" are invalid values and can be removed.
- **Correct wrong structure:** Values that don't follow a defined structure can be removed. E.g. In a data set containing pin codes of Indian cities, a pin code of 12 digits would be an invalid value and needs to be removed. Similarly, a phone number of 12 digits would be an invalid value.
- **Validate internal rules:** If there are internal rules such as a date of a product's delivery must definitely be after the date of the order, they should be correct and consistent.

Checklist for filtering data:

- **Deduplicate data:** Remove identical rows, remove rows where some columns are identical
- **Filter rows:** Filter by segment, filter by date period to get only the rows relevant to the analysis
- **Filter columns:** Pick columns relevant to the analysis
- **Aggregate data:** Group by required keys, aggregate the rest

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